

Open Cities Hackathon: Round 1 Brief

This brief was written for the TraceGraph eval suite and is released into the public domain. The event, organisers and prizes are fictional.

Theme

Build a working prototype that helps a mid-sized city run its public services more fairly and more cheaply, using open data and several cooperating software agents. Teams of two to five people have 48 hours. Every entry must run on open data that anyone can download, and every agent in the system must explain what it did in plain language.

Tracks

Track 1, Clean Transit. Cut waiting times on buses and trams. Use the city timetable feed (GTFS), live vehicle positions and ridership counts to find the routes where riders wait longest, and propose timetable changes a planner could adopt the next day.

Track 2, Safe Streets. Reduce road injuries near schools. Combine collision records, speed surveys and school locations to rank the most dangerous crossings, and draft a short report a council meeting could vote on.

Track 3, Fair Housing. Help renters find out whether a rent increase is legal. Read the local rent rules, compare an offered rent with the registered rents nearby, and write a clear letter the renter can send to the landlord.

Rules

- All code must be written during the event; libraries and public models are allowed.
- Personal data must never leave the team machine, and demo data must be synthetic.
- Each team submits a public repository, a three-minute video and a one-page design note.
- A system must use at least two agents that hand work to each other, with logs of every hand-off.
- Paid APIs may be used only if the demo still runs, more slowly, without them.

Judging criteria

- Impact (30 points): how many residents the idea helps, and how much.
- Working demo (25 points): the judges run it themselves on a fresh laptop.
- Use of agents (20 points): the agents divide the work sensibly and explain their steps.
- Openness (15 points): open data, open code, and instructions another city could follow.
- Presentation (10 points): the video and the design note are clear and honest about limits.

Timeline

Friday 18:00 kick-off and data briefing. Saturday 12:00 mentor check-in, where each team shows a one-slide plan. Sunday 14:00 code freeze. Sunday 15:00 to 17:00 judging. Sunday 17:30 results. The top two teams in each track go through to Round 2, a four-week pilot with a partner city.

Constraints and technology

Prototypes must run on a single laptop with 16 GB of memory and no GPU. The city data portal offers CSV and JSON downloads and a rate-limited API of 60 requests a minute. Teams should

expect missing values and duplicate rows, and say how they handled them in the design note.